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THE QUALITY MANAGEMENT THEORY

Supplier Quality Management + Prevention-Based Quality

Quality should not depend only on detecting defective components after they arrive. Supplier processes should be monitored and improved proactively so that variation and potential nonconformities are controlled before they affect the final system.

THE INDUSTRIAL QUALITY PROBLEM

Illustrative Quality Scenario — Not an Actual ASML Incident

A critical component supplied for a high-precision semiconductor manufacturing system meets the current specification during incoming inspection.

However, supplier process data shows:

- Increasing process variation
- Fluctuating process behaviour
- Borderline process capability
- Recurring nonconformities over time

The immediate problem can be contained, but simply replacing defective components does not eliminate the underlying risk.

Core Question: How can we detect and control supplier-process risk before it becomes a recurring product-quality problem?

QUALITY TOOLS USED

SPC → Process Capability → Trend Analysis → RCA → Supplier Audit → Corrective Action → Verification → FMEA

SPC: Detect process variation & instability.

Capability Analysis: Determine if process meets requirements consistently.

RCA: Identify why variation/nonconformity occurs.

Corrective Action: Address underlying cause.

Effectiveness Verification: Confirm improvement sustainability.

PROPOSED QUALITY SOLUTION

Supplier Process → Detect Variation → Capability Assessment → RCA → Corrective Action → Verify → Continuous Monitoring

1. **Detect:** Monitor supplier process data instead of relying only on incoming inspection.
2. **Analyse:** Investigate variation, stability, and process capability.
3. **Identify Root Cause:** Determine if issues stem from parameters, equipment, material, method, or measurement.
4. **Improve:** Implement corrective/preventive actions collaboratively with the supplier.
5. **Verify:** Use data to confirm capability and process stability improvements.
6. **Prevent Recurrence:** Continue monitoring and update supplier risk assessments.

WHY THIS MATTERS IN HIGH-PRECISION EQUIPMENT

Semiconductor manufacturing equipment depends on extreme precision components and a complex global supplier ecosystem. Unchecked variation introduces severe risk:

Supplier Variation → Component Risk → System Performance Risk

Proactive supplier quality management identifies quality risks early rather than depending solely on final inspection. (Note: Based solely on publicly available information; does not reflect internal ASML procedures.)

EXPECTED QUALITY IMPACT

- &d5; Supplier-related defects
- &d5; Process variation
- &d5; Recurring nonconformities
- &u5; Component reliability & consistency

Better supplier performance visibility + earlier risk detection + stronger prevention.

“A supplier part can meet today’s specification and still carry tomorrow’s quality risk if process variation is not understood and controlled.”